Diego A. Lopez G.

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Education

PhD **University of Kentucky**, Electrical Engineering — Lexington, KY
- Advisor: Professor Dan M. Ionel — *May 2029*
- GPA: 4.0/4.0

BSc **Universidad de La Sabana**, Mechanical Engineering — Chia, Colombia
- Excellence Scholarship recipient — Mar 2024
- GPA: 3.5/4.0 (equiv. excellent)

Experience

SPARK Lab, University of Kentucky, Research Assistant — Lexington, KY
- Contributed to electric machine projects sponsored directly by NASA and industry — Aug 2024 – present
- Designed, manufactured, and assembled an axial-flux rotor with 0.5 MPa of magnetic attraction — 2 years 4 months
- Produced 3D CAD models, GD&T manufacturing drawings, and assembly videos
- Performed parametric electromagnetic analysis and optimization in Ansys Maxwell, and structural FEA in Ansys Workbench
- Tested different motor topologies implementing motor control with dSpace and Simulink
- Processed simulation data in Python and MATLAB to produce paper- and patent-ready graphs

Stanley and Karen Pigman College of Eng., University of Kentucky, Teaching Assistant — Lexington, KY
- Guided engineering students from diverse backgrounds through complex engineering concepts — Aug 2024 – present
- *Electromechanics and Electromagnetics:* First-principles theoretical foundations in electric machines and engineering electromagnetics — 2 years 4 months
- *First-Year Engineering:* Microcontroller-based projects with MATLAB, CAD, and 3D printing

Projects

Project Polaris — 2022 – 2023
- Designed and built an autonomous atmospheric balloon probe prototype for Titan exploration in a student space program
- Led an international, multidisciplinary team of 50+ members
- Developed instrumentation and actuation systems with Raspberry Pi and Arduino
- Coauthored research papers: IAC 2022 and AIAA SciTech 2024

Ultra-Efficient EV — 2019 – 2024
- Designed and built an ultra-efficient battery-electric vehicle for Shell Eco-marathon
- Developed and tested a BLDC motor speed controller, from KiCad PCB design to manual assembly
- Honorable mention for innovative technical design at Shell Eco-marathon Brazil 2022
- Modeled parts in Inventor; manufactured carbon fiber, 3D-printed, and CNC-machined parts
- Solved engineering problems as part of a fast-paced, collaborative team

Skills and Relevant Coursework

Soft skills: Excellent written and verbal communication, result-oriented, and strong interpersonal skills with demonstrated effectiveness in a fast-paced team environment

Software: Autodesk Inventor and Fusion 360, Ansys Maxwell and Workbench (FEA), MATLAB, Python, KiCad

Prototyping: Manufacturing drawings with GD&T, mechanical fixture design for motor prototypes, 3D printing, PCB design and assembly, motor testing and control, microcontroller systems

Ph.D. coursework: Electric Machines Computer Aided Eng. (A), Machine Learning for Motor Design and Analysis (A), Electromagnetic Fields (A), Electric Vehicles and Aircraft (A), Mechanical Design with Finite Element Methods (A)

Research Areas: Electric machine design and optimization, axial-flux motors, PCB windings, electric aircraft

Publications

PM-Free Axial-Flux AC Motor with Reluctance Rotor and Ferromagnetic-Core Stator with PCB DC Excitation for Renewable Energy Systems
Lopez-Guerrero, D. A., Vatani, M., Ionel, D. M.
(DOI pending, ICRERA 2026)

Halbach-Mallinson Array Rotor with Variable-Length PMs for Axial-Flux Machines
Lopez-Guerrero, D. A., Vatani, M., Goode-Kulchar, S., Ionel, D. M.
(DOI pending, ECCE 2026)

Fault Detection and Mitigation in a Multi-Stage Coreless Axial-Flux PM Motor System
Deponti, M., Vatani, M., ***Lopez-Guerrero, D. A.***, Perini, R., Ionel, D. M.
(DOI pending, ECCE 2026)

Electrical Fault Tolerance of a Multi-Stage Axial-Flux PM Drivetrain for Electrified Propulsion
Deponti, M., Lewis, D. D., ***Lopez-Guerrero, D. A.***, Vatani, M., Perini, R., Ionel, D. M.
10.1109/ITECEATS66641.2026.11593010 (ITEC 2026)

Multi Electric Machines with Series and Parallel Electromechanical Combinations for Aircraft
Stewart, D. R., Lewis, D. D., Vatani, M., ***Lopez-Guerrero, D. A.***, Ionel, D. M.
10.1109/ECCE58356.2025.11260309 (ECCE 2025)

Mitigation of PM Eddy Current Losses Using Offset Windings in Coreless AFPM Machines
Vatani, M., ***Lopez-Guerrero, D. A.***, Eastham, J. F., Ionel, D. M.
10.1109/ITEC63604.2025.11098002 (ITEC 2025)

Selected Honors

- Best Student Demo Award, IEEE International Electric Machines & Drives Conference 2025
- Winner of the 2021 Sabanahack hackathon, awarded by Prodigious and Universidad de La Sabana
- Winner of the Excellence Scholarship awarded by the Universidad de La Sabana College of Engineering (2019–2023)
- Scored in the top 1% nationally on standardized exams for high school (2018) and university graduates (2023)
- Certified C1 English level (IELTS 7.5; Duolingo English Test 145)

References

Professor Dan M Ionel, PhD, MEng, IEEE Fellow
L. Stanley Pigman Chair in Power, University of Kentucky
Honorary Professor, University College London (UCL)
Director, SPARK Lab and PEIK Institute
University of Kentucky, dan.ionel@uky.edu